Early Prediction of Vasopressor Initiation

Using MIMIC-IV EHR Data and Machine Learning

Comparing demographics and laboratory results with early blood-pressure levels and trends

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MIMIC-IV • A controlled comparison of demographics/labs vs early blood-pressure information

Q: Can the first six hours of ICU data predict vasopressor initiation in the next 24 hours?

- Binary classification using routinely recorded EHR data
- Three ML algorithms × three feature sets
- Held-out evaluation and hyperparameter-tuning bonus

Main finding: enhanced Gradient Boosting test AUPRC = 0.400 vs 0.318 baseline.

Why this prediction task matters

Clinical motivation and hypothesis

- Vasopressors are administered to support circulation in selected critically ill patients.
- Earlier risk estimates could inform clinician awareness and patient monitoring.
- Hypothesis: early BP levels and temporal summaries add information beyond demographics and laboratory results.
- This is prediction of observed treatment initiation, not diagnosis of shock or a treatment recommendation.

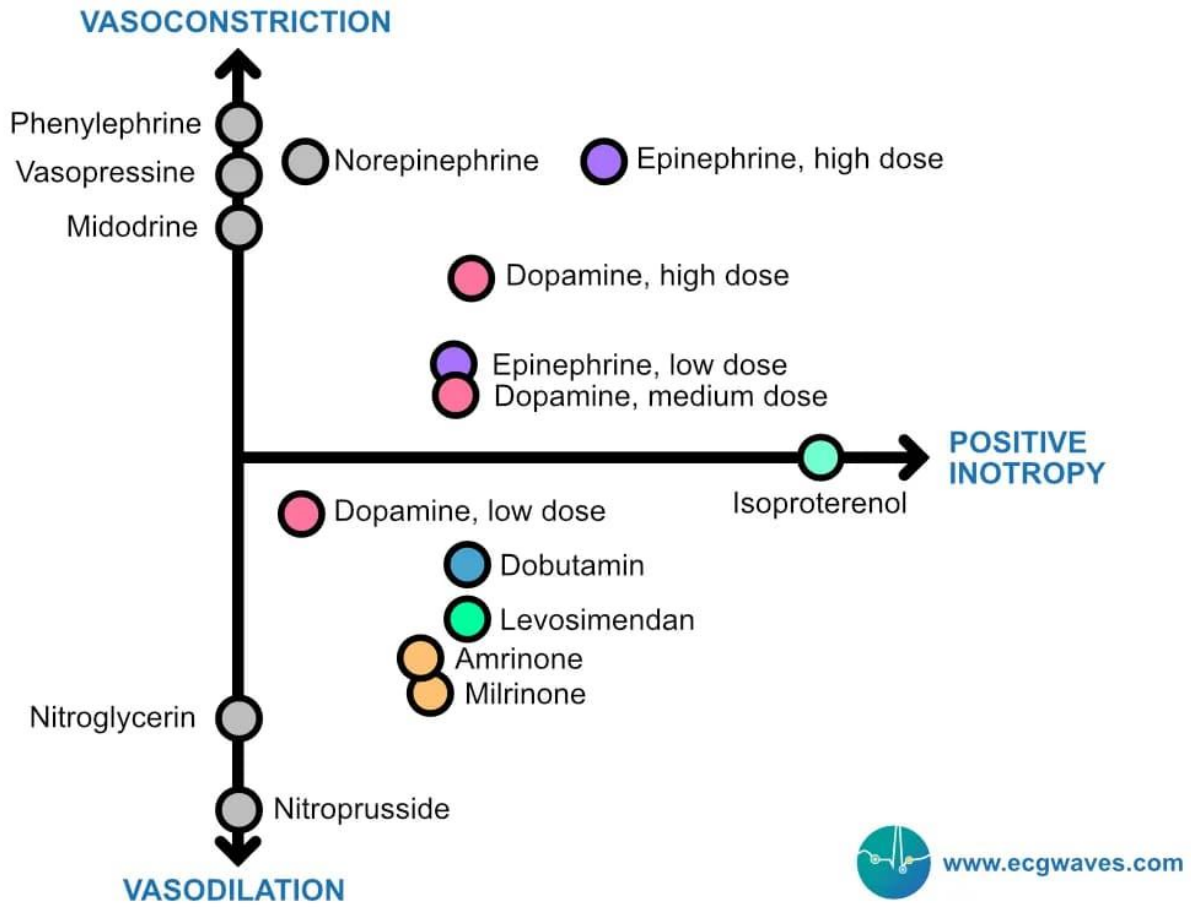


Figure: Overview of vasopressors and inotropes

Prediction design and cohort eligibility

A fixed six-hour landmark prevents future measurements from entering the feature set

Time	Study role	Description
0–6 hours	Observation	Demographics, admission fields, labs, BP
At hour 6	Prediction landmark	Patients still eligible; no initial vasopressor use
>6–30 hours	Binary outcome	New documented vasopressor initiation

- Adult patients; first ICU stay per patient; observed in ICU through hour 30.
- Exclude documented vasopressor administration in hours 0–6.
- Eight audited MIMIC-IV medication item IDs used for labeling.

Vasopressor label audit

MIMIC-IV ICU inputevents • verified administered formulations

Medication / formulation	Item ID	Valid administration rows
Norepinephrine	221906	459,800
Phenylephrine	221749	209,374
Phenylephrine (50/250)	229630	61,013
Vasopressin	222315	37,163
Epinephrine	221289	31,495
Dopamine	221662	18,085
Phenylephrine (200/250)	229632	12,840
Epinephrine.	229617	233

Excluded historical phenylephrine _OLD_1 label: zero qualifying administrations.

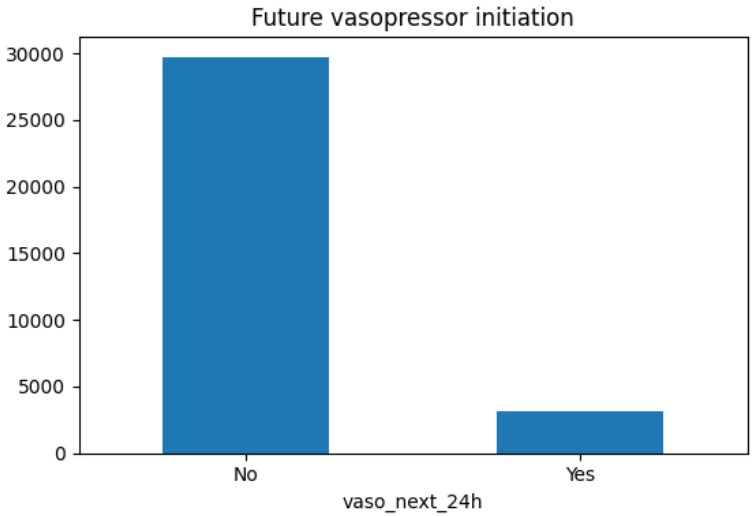
Final analytical cohort

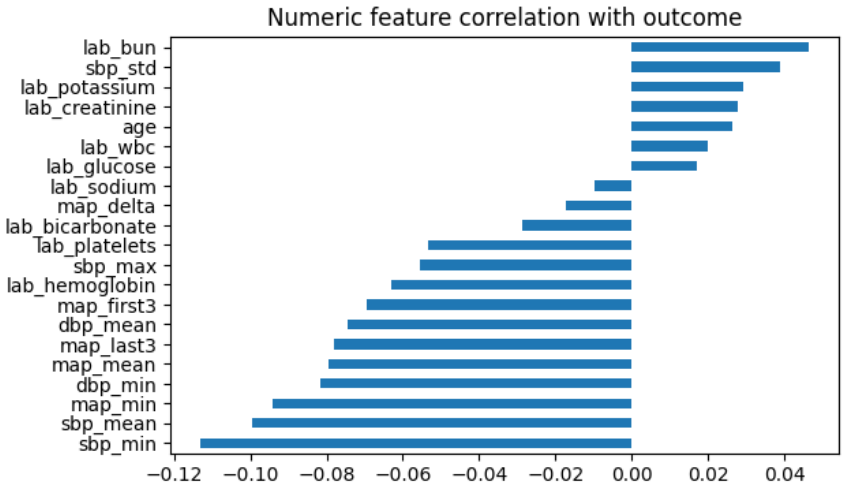
Study cohort and outcome distribution

32,878
Eligible first ICU stays

3,160
Future initiations

9.61%
Event prevalence





Train 21,041 (64%) • Validation 5,261 (16%) • Test 6,576 (20%)

Cohort heterogeneity across ICU units

Observed new-initiation rates differ meaningfully by care unit

Care unit	Eligible stays	Events	Rate
Medical ICU	6,226	610	9.8%
Surgical ICU	5,075	349	6.9%
Medical/Surgical ICU	5,021	329	6.6%
Trauma SICU	3,925	367	9.4%
Coronary Care Unit	3,820	253	6.6%
Cardiac Vascular ICU	3,748	1,092	29.1%
Neuro Intermediate	3,351	53	1.6%

Interpretation

Care unit may capture differences in patient case mix and treatment practices.

Stepdown/intermediate labels remain within the extracted cohort.

Feature engineering: three nested feature sets

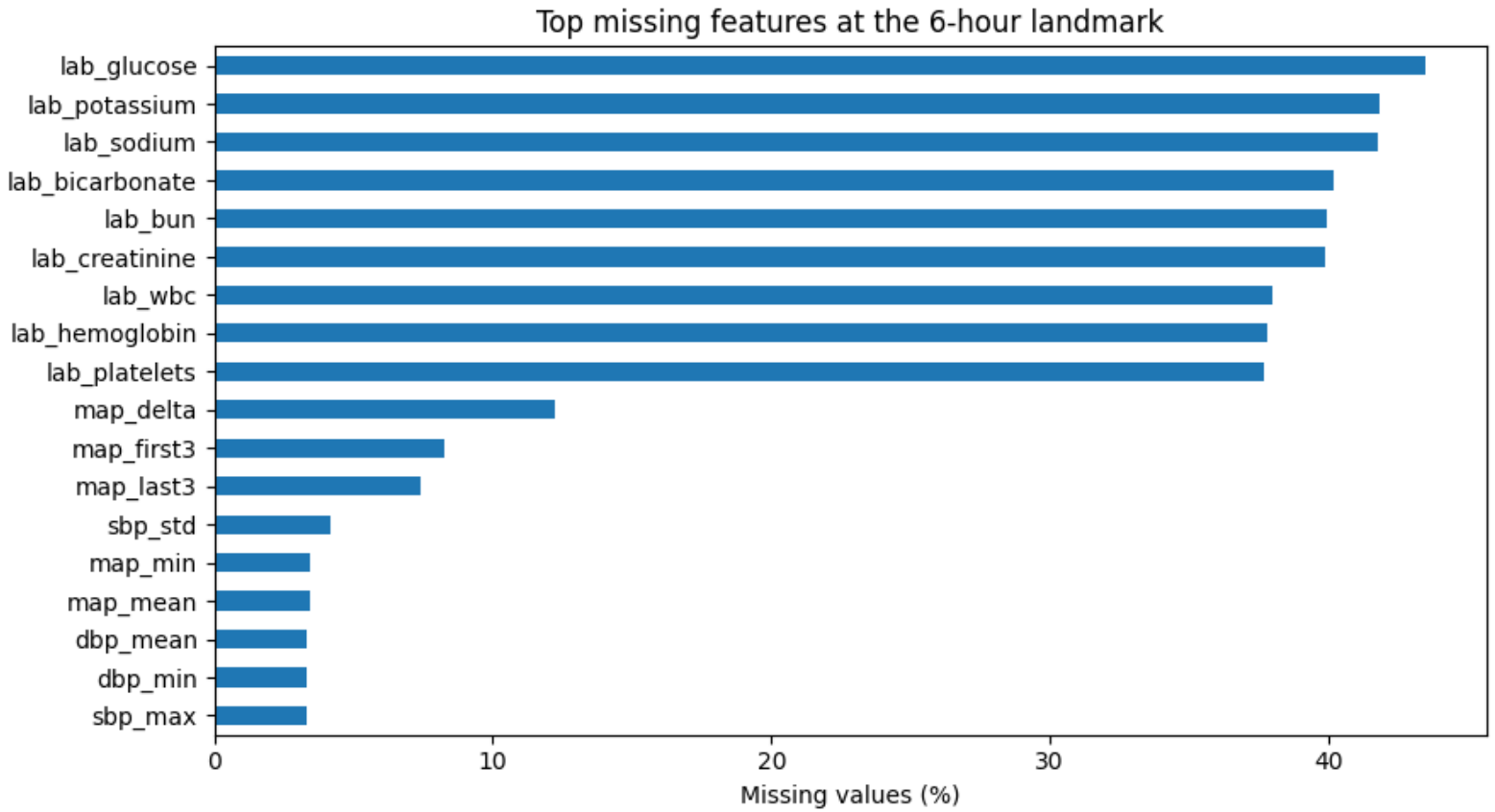
Same patients, algorithms, and splits; only the available information changes

Feature set	Included predictors
A · Baseline	Age, sex, admission attributes, 9 first-six-hour labs
B · BP levels	A + SBP mean/min/max, DBP mean/min, and MAP mean/min
C · BP trends	B + SBP variability (SD), MAP averages first 3h / last 3h, MAP change in averages

- Baseline admission features include first ICU care unit and insurance type.
- Nine first-six-hour labs: WBC, hemoglobin, platelets, sodium, potassium, creatinine, BUN, glucose, bicarbonate.
- Missing lab values are imputed within the modeling pipeline, not using held-out labels.

Data completeness and exploratory analysis

First-six-hour labs are substantially more incomplete than BP measurements



Missingness

Labs: ~37.7–43.5% missing.

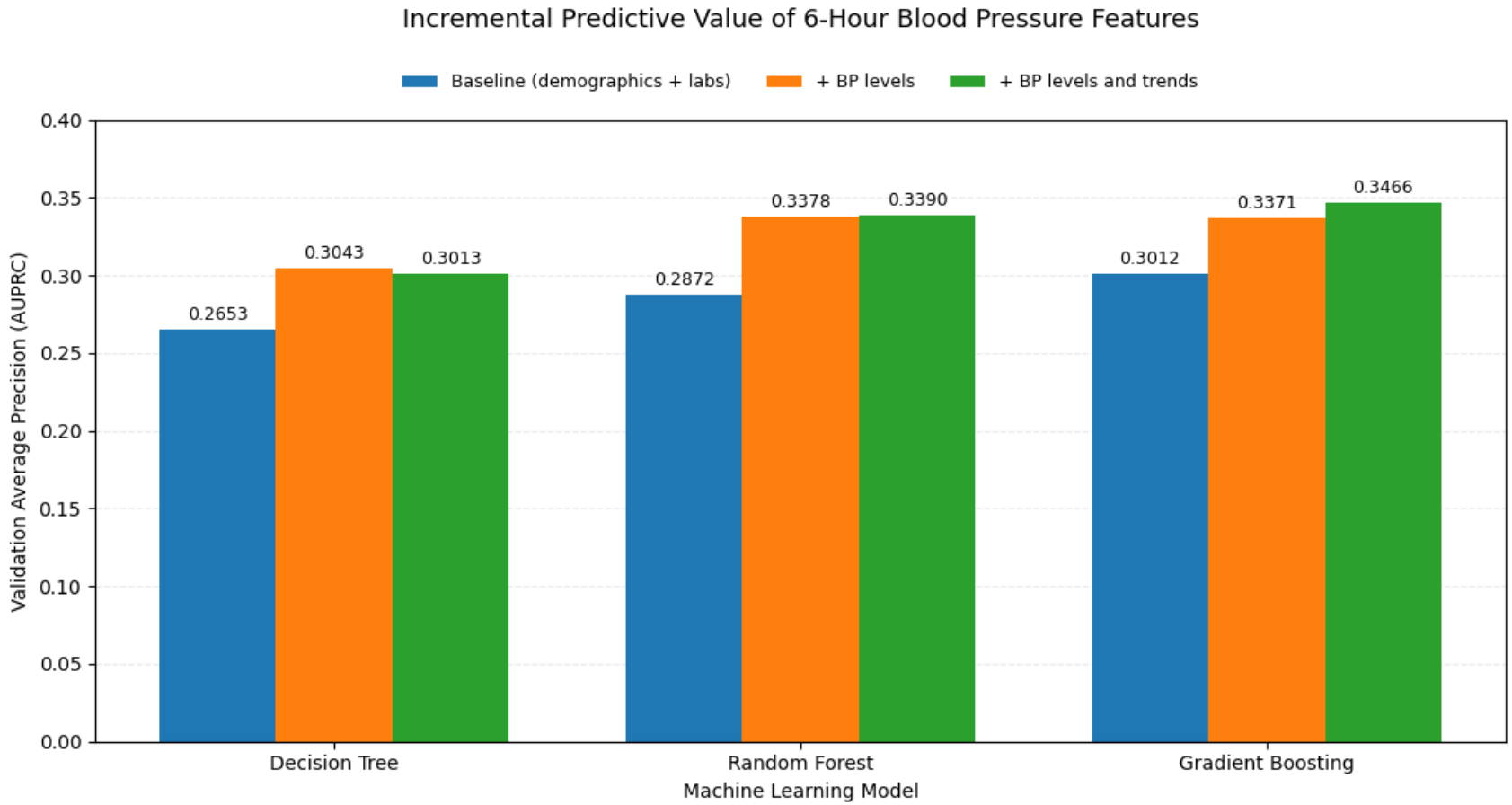
BP means/minima: ~3.3–3.4% missing.

MAP change: ~12.2% missing.

Missingness may reflect clinical testing and charting patterns.

Model selection on the validation cohort

Three algorithms × three feature sets; metric prioritized: AUPRC



Best validation model: Gradient Boosting + BP trends (AUPRC 0.3466).

Ablation: where does the BP improvement come from?

Validation AUPRC across three feature sets

Model	Baseline	BP levels	BP levels + trends
Decision Tree	0.2653	0.3043	0.3013
Random Forest	0.2872	0.3378	0.3390
Gradient Boosting	0.3012	0.3371	0.3466

- All three algorithms improve from the baseline to the full BP feature set.
- Most of the improvement comes from BP levels; additional trends yield smaller, model-dependent gains.
- Gradient Boosting: +0.0359 from levels, then +0.0095 from added temporal summaries.

Final held-out comparison: Gradient Boosting

Identical 6,576-person test cohort; ranking metrics are threshold independent

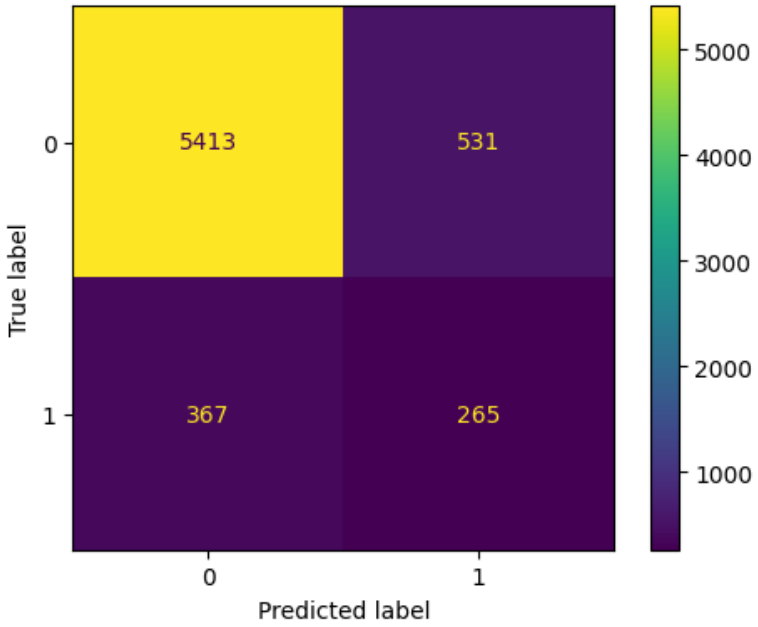
Feature set	Test AUROC	Test AUPRC
Baseline demographics + labs	0.7245	0.3182
+ BP levels	0.7665	0.3883
+ BP levels and trends	0.7692	0.3996

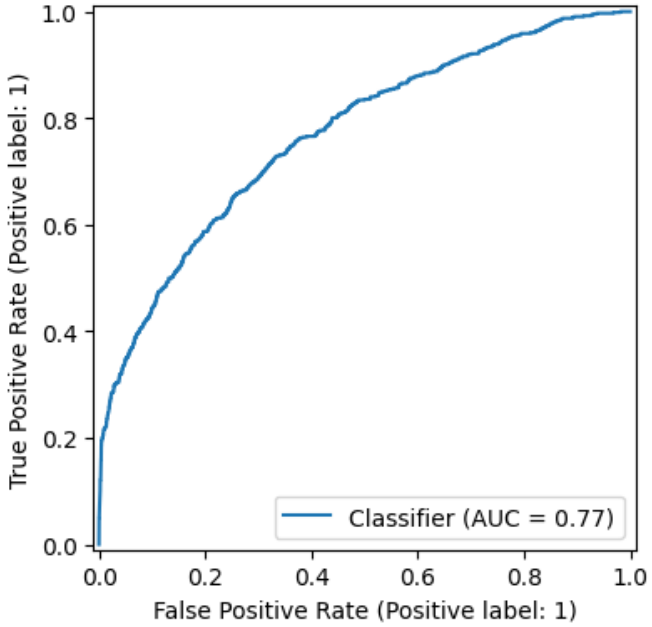
Full model vs baseline: +0.0814 AUPRC (+25.6% relative).

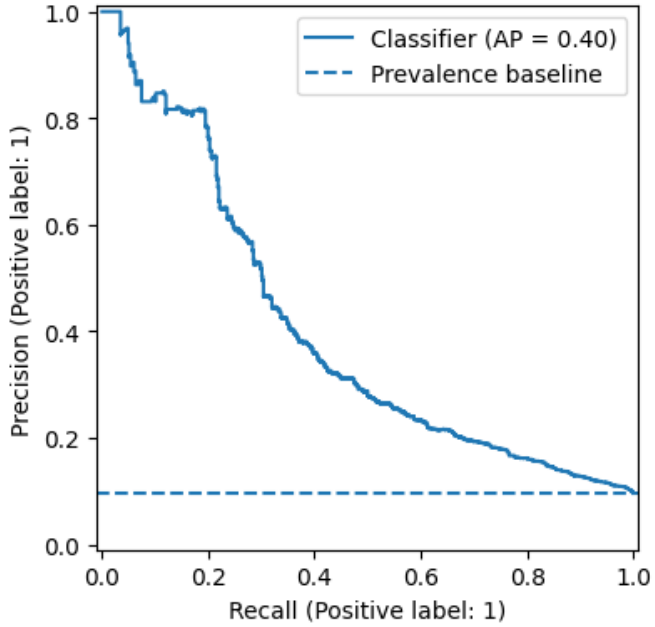
Paired bootstrap 95% CI for AUPRC gain: +0.0553 to +0.1034.

Final test diagnostics

Selected Gradient Boosting; classification threshold chosen on validation F1





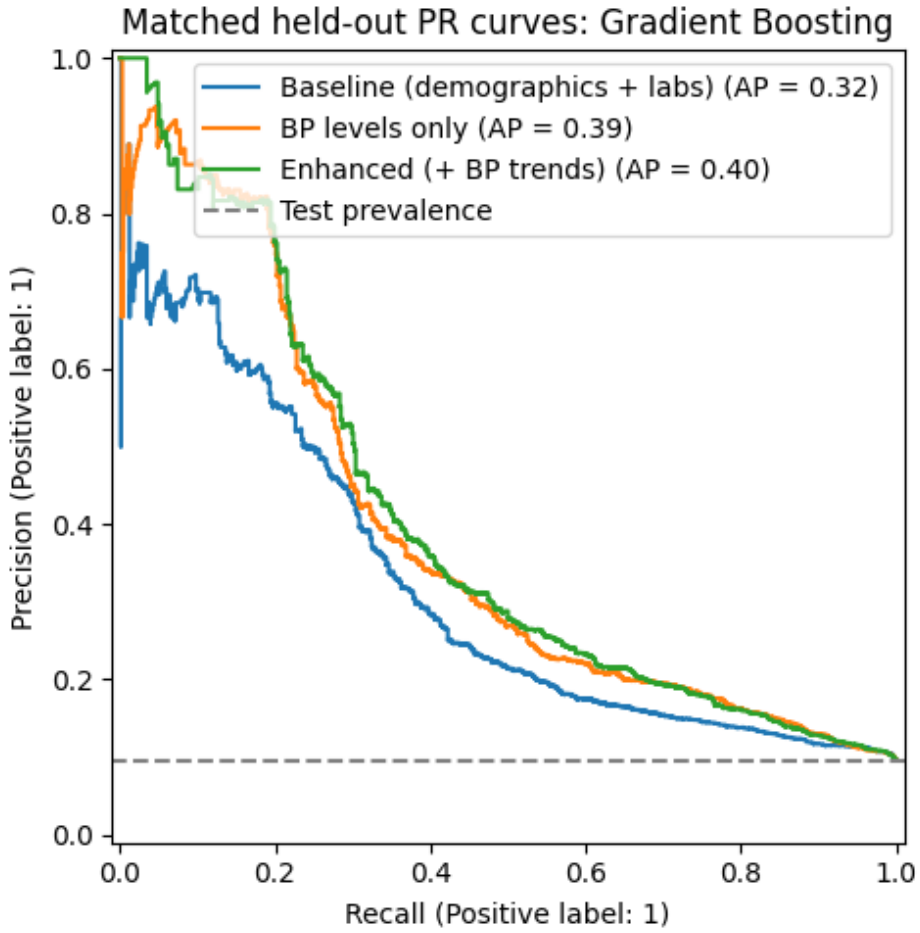


AUROC	AUPRC	Precision	Recall	F1	Threshold
0.769	0.400	0.333	0.419	0.371	0.151

At this threshold, recall is 41.9% and precision is 33.3%; not a clinical decision rule.

Uncertainty in performance improvement

Paired bootstrap of matched held-out predictions



Observed ΔAUPRC

+0.0814

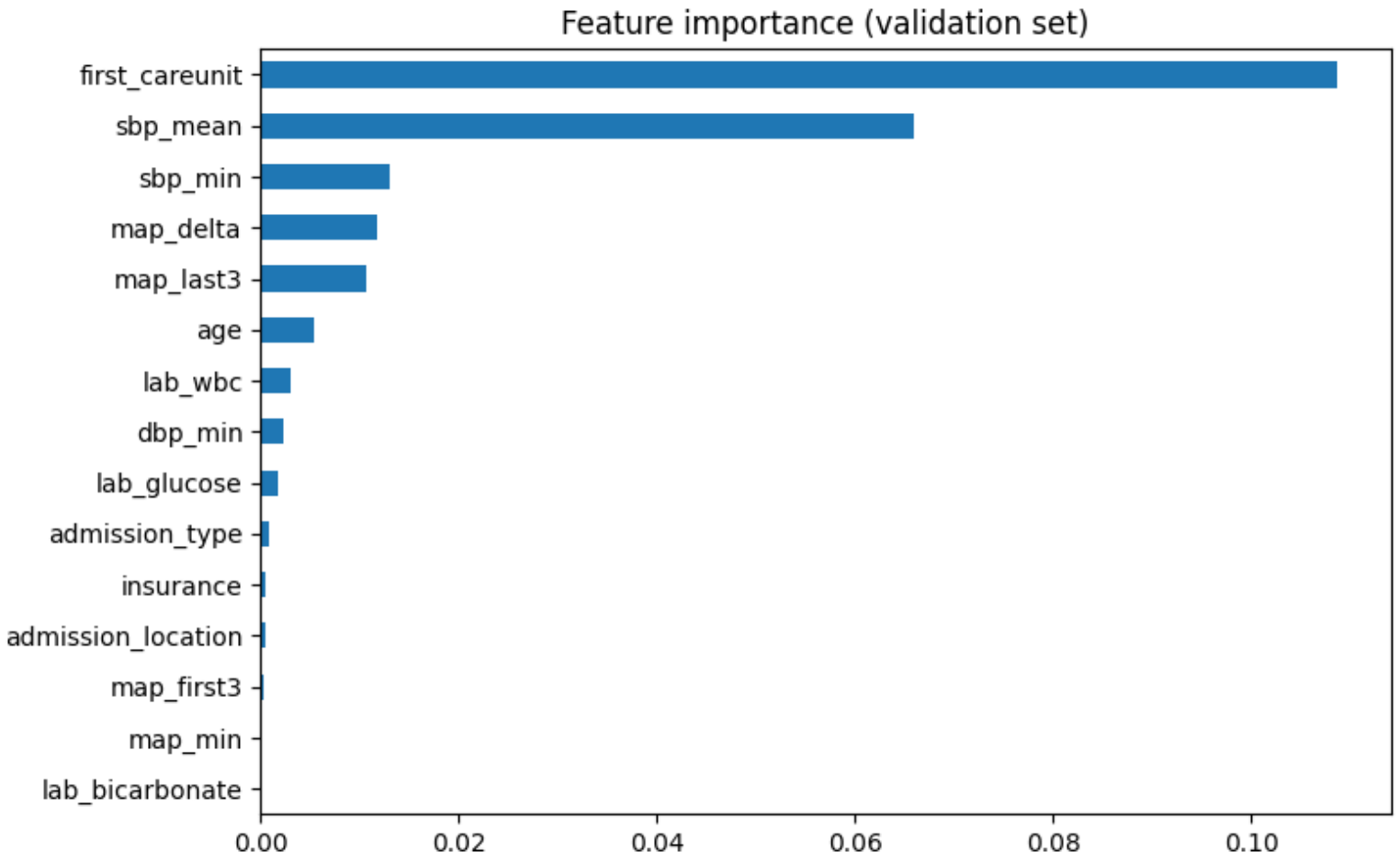
95% percentile interval

[+0.0553, +0.1034]

Positive interval supports improvement in this selected test cohort.

What the model relies on

Validation-set permutation importance, measured as the drop in AUPRC



Top signals

First care unit: 0.109

Mean SBP: 0.066

Minimum SBP: 0.013

MAP change: 0.012

Late MAP: 0.011

Predictive association ≠ causation.

Bonus: hyperparameter tuning

GridSearchCV for Random Forest: 8 configurations × 3-fold CV

Model	Test AUROC	Test AUPRC	F1 (model-specific threshold)
Selected enhanced Gradient Boosting	0.769	0.400	0.371 (0.151)
Tuned Random Forest	0.768	0.394	0.369 (0.5)

- Best RF parameters: 200 trees, unrestricted max depth, minimum 10 samples per leaf.
- Best training-CV AUPRC: 0.371; tuned RF validation AUPRC: 0.341.
- Hyperparameter tuning improved Random Forest validation AUPRC only marginally, from 0.339 to 0.341 (+0.002), demonstrating limited benefit from optimization.
- The tuned RF was competitive but did not exceed the selected Gradient Boosting test AUPRC (0.394 vs 0.400).

Limitations and interpretation

Important constraints on the strength and generalizability of the results

- Selection bias: requiring ICU observation through hour 30 uses future stay duration and excludes early exits.
- The outcome is documented vasopressor administration, not a diagnosis of shock or treatment need.
- Care-unit effects may reflect case mix and clinician treatment practices.
- Lab missingness, measurement frequency, and fragmented infusion records may introduce bias or mislabeling

Conclusion and key takeaways

Answering the project question with evidence from the executed notebook

- The full BP-enhanced Gradient Boosting model achieved test AUPRC 0.400 vs 0.318 for demographics/labs only.
- The +0.0814 AUPRC improvement had a paired bootstrap 95% interval of +0.0553 to +0.1034.
- BP levels drove most of the gain; temporal summaries added a smaller incremental improvement.
- Model performance is encouraging but sensitivity, precision, selection bias, and site specificity limit clinical application.

Takeaway: early BP information adds meaningful predictive signal in this retrospective cohort.